INCH-POUND
A-A-60035
5 July 2022

COMMERCIAL ITEM DESCRIPTION

BOLTS, CADMIUM AND HEXAVALENT-CHROMIUM FREE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1 SCOPE. This commercial item description (CID) covers bolts which shall not contain cadmium or hexavalent-chromium in their finishes.

2 CLASSIFICATION

2.1 Types.

Type I	-	NAS1251 - Bolt, Ring, Swivel
Type II	-	NAS1297 - Bolt, Shoulder-Hexagon Head
Type III	-	NAS6603 THRU NAS6620 - Bolt, Tension, Hex Head, Close Tolerance, Alloy Steel, Long Thread, Reduced Major Dia., Self-Locking and nonlocking, 160 KSI Ftu
Type IV	-	NASM3 THRU NASM20 - Bolt, Machine, Aircraft
Type V	-	NASM21098 - Bolt, Self-Locking, Steel, 160 KSI Ftu, 250 °F, 12 Point, External Wrenching (Externally Wrenching Cap Screws)
Type VI	-	NASM21250 - Bolt, Tension, Steel, External Wrenching, Flanged, 12-Point, 180 KSI Ftu, 450 ºF
Type VII	-	NASM35751 - Bolt, Square Neck, Round Head (Carriage), Steel, Cadmium Or Zinc Plated, UNC-2A

3 SALIENT CHARACTERISTICS.

3.1 Materials. Bolts shall be manufactured as specified in the Aerospace Industry Association (AIA) documents listed in 2.1 by Type, except the plating/finishes shall not contain cadmium and hexavalent-chromium as specified herein.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 5306

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3.2 Dimensions. Dimensions for all designated types of bolts apply after plating/finishing. Refer to each individual NAS/NASM specification listed in Table 2.1 for detailed part geometry and tolerance requirements.

3.3 Finish/plating. Finish/plating types for each type of bolt shall be designated as in 7.1.1. The finishes/plating options for the bolts are listed below:

a. Aluminum shall be chemically treated in accordance with MIL-DTL-5541, Type II, Class 3, or MIL-DTL-81706, Type II, Class 3; or anodized in accordance with MIL-PRF-8625, Type IC or IIB, Class 1. Sealers (clear or any color) shall not contain hexavalent chromium.
b. Carbon and Alloy steel shall be Zinc-Nickel plated in accordance with MIL-PRF-32647, Type K16.
c. CRES 340 and CRES 316 shall be cleaned, descaled and passivated in accordance with SAE AMS 2700 Method 1, Type 6 or 7, or Method 2, post treatment is prohibited.

4 REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation (FAR), or the sustainable acquisition of EO 13834; as applicable.

5 PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Responsibility for inspection. The contractor is responsible for the performance of all inspections.

5.3 Contractor certification. The contractor shall certify and maintain substantiating evidence the product offered meets the salient characteristics of this commercial item description and conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to perform or witness any of the inspections it deems necessary to ensure supplies and services conform to prescribed requirements.

6 PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7 NOTES

The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

7.1 Table. The following Tables indicate new finish/plating PIN's which are Cadmium and hexavalent-chromium free for the designated bolts.

7.1.1 Cadmium and hexavalent-chromium free PIN codes. Cadmium and hexavalent-chromium free part numbers shall be as identified by the examples herein and in conjunction with AIA/NAS/NASM dash numbers listed in the respective specifications (see 2.1). The following finish/plating (**cadmium and hexavalent-chromium free**) codes are given:

QX - Steel – Zinc-Nickel, MIL-PRF-32647, Type K16
P5 - CRES – Passivate SAE AMS2700, Method 1 Type 6 or 7, or Method 2, no post treatment
J - AL - Chemically treated per MIL-DTL-5541, Type II, Class 3 or MIL-DTL-81706, Type II, Class 3
K - Aluminum – Anodize per MIL-PRF-8625, Type IC or IIB, Class 1
H - No finish/plating

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TABLE I – Type I (-1) - Bolt, ring, swivel.

A-A-60035 Part Number (Zinc-Nickel Plated)	Reference Dash Number (NAS1251)
AA60035-1-2QX	NAS1251-2
AA60035-1-5QX	NAS1251-5
AA60035-1-A10QX	NAS1251A10
AA60035-1-A25QX	NAS1251A25

Example of part numbers ("D" in the part number = Drilled hole in thread end per NAS1251):

AA60035-1-5D8QX = a NAS1251 - Bolt, ring, 5,000 LBF rated capacity, drilled thread, 0.500 grip length, Zinc-Nickel Type K16 plated.

AA60035-1-A10QX = a NAS1251 - Bolt, ring, 10,000 LBF rated capacity, without drilled thread, threaded to shoulder, Zinc-Nickel Type K16 plated.

TABLE II – Type II (-2) - Bolt, shoulder-hexagon head.

A-A-60035 Part Number (Zinc-Nickel Plated)	Reference Dash Number (NAS1297)
AA60035-2-06*#QX	NAS1297-06*#
AA60035-2-3*#QX	NAS1297-3*#
AA60035-2-4*#QX	NAS1297-4*#
AA60035-2-5*#QX	NAS1297-5*#
AA60035-2-6*#QX	NAS1297-6*#
AA60035-2-7*#QX	NAS1297-7*#
AA60035-2-8*#QX	NAS1297-8*#
AA60035-2-10*#QX	NAS1297-10*#

* Indicates:

"-" = undrilled
"D" = Drilled Shank
"H" = Drilled Head
"DH" = Drilled Shank and Head

Indicates Grip Length as in Table II of NAS1297

Example of part numbers:

AA60035-2-06-4QX = a NAS1297 – Shoulder bolt with a hexagon head, alloy steel, 0.1380-32 UNJF-3A, undrilled, 0.260 grip length, Zinc-Nickel Type K16 plated.

AA60035-2-5D8QX = a NAS1297 – Shoulder bolt with a hexagon head, alloy steel, 0.3125-24 UNJF-3A, drilled shank, 0.510 grip length, Zinc-Nickel Type K16 plated.

AA60035-2-10DH10QX = a NAS1297 - Shoulder bolt with a hexagon head, alloy steel, 0.6250-18 UNJF-3A, drilled shank and head, 0.635 grip length, Zinc-Nickel Type K16 plated.

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TABLE III – Type III (-3) - Bolt, tension, hex head, close tolerance, alloy steel, long thread, reduced major dia., self-locking and nonlocking, 160 KSI Ftu, Zinc-Nickel Plated per MIL-PRF-32647, Type K16.

A-A-60035 Basic Part Number (Alloy Steel, Zinc-Nickel Plated)	Reference Dash Number (NAS6603 THRU NAS6620)
AA60035-3-03*QX	NAS6603
AA60035-3-04*QX	NAS6604
AA60035-3-05*QX	NAS6605
AA60035-3-06*QX	NAS6606
AA60035-3-07*QX	NAS6607
AA60035-3-08*QX	NAS6608
AA60035-3-09*QX	NAS6609
AA60035-3-10*QX	NAS6610
AA60035-3-12*QX	NAS6612
AA60035-3-14*QX	NAS6614
AA60035-3-16*QX	NAS6616
AA60035-3-18*QX	NAS6618
AA60035-3-20*QX	NAS6620

* Feature Code(s) after Basic Part Number as follows:

"D" after size code = drilled hole in thread end
"-" = undrilled, (DO NOT USE with "L" or "P" code.)
"H" after size code = drilled head
"P" after size code = self-locking bolt with patch type locking element type
"L" after size code = self-locking bolt with locking element type (optional)

Add grip dash number indicates grip in 0.0625 increments (see NAS6603 THRU NAS6620).

For oversized bolts, same referenced dash number requirements apply.

Examples of part number:

AA60035-3-04-10QX = a NAS6604 - Bolt, 0.2500-28 thread, 0.625 grip, undrilled, nonlocking, Zinc-Nickel Type K16 plated.

AA60035-3-04H10QX = a NAS6604 - Bolt, 0.2500-28 thread, 0.625 grip, drilled head, undrilled thread, nonlocking, Zinc-Nickel Type K16 plated.

AA60035-3-04DH10QX = a NAS6604 - Bolt, 0.2500-28 thread, 0.625 grip, drilled thread, drilled head, nonlocking, Zinc-Nickel Type K16 plated.

AA60035-3-04L10QX = a NAS6604 - Bolt, 0.2500-28 thread, 0.625 grip, undrilled, self-locking (locking type optional), Zinc-Nickel Type K16 plated.

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TABLE IV – Type IV (-4) - Bolt, machine, aircraft.

A-A-60035 Basic Part Number (See below for material/finish codes)	Reference Basic Part Number (NASM3 THRU NASM20)
AA60035-4-3*	AN3
AA60035-4-4*	AN4
AA60035-4-5*	AN5
AA60035-4-6*	AN6
AA60035-4-7*	AN7
AA60035-4-8*	AN8
AA60035-4-9*	AN9
AA60035-4-10*	AN10
AA60035-4-12*	AN12
AA60035-4-14*	AN14
AA60035-4-17*	AN17
AA60035-4-18*	AN18
AA60035-4-20*	AN20

* Feature Code(s) after Basic Part Number as follows:

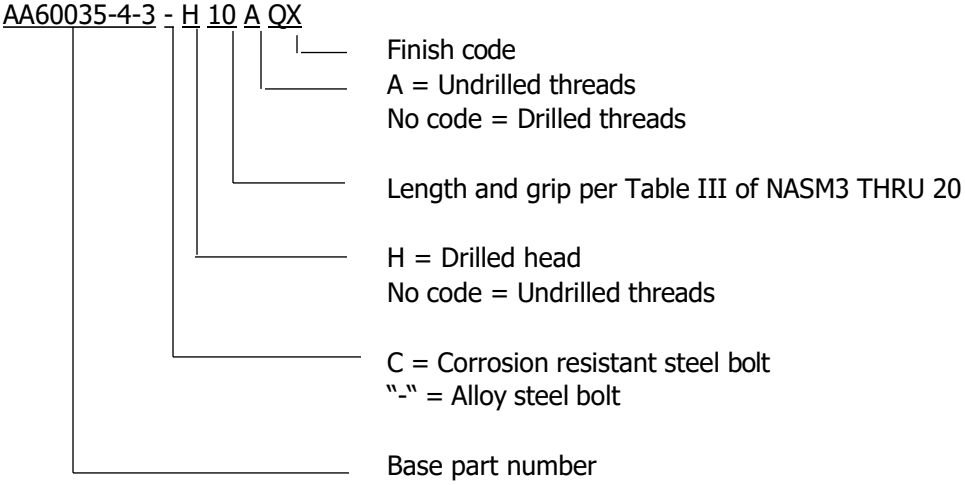
Add "-" for Steel (non-corrosion resistant steel bolt)
Add "C" for Corrosion-resistant steel bolt

Add "H" for Drilled Head
No letter for Undrilled Head

Add "length and grip number" code (see NASM3 thru NASM20):

Add "A" for Undrilled Thread
No letter for Drilled Thread

Add plating/finish code:
QX = Zinc-Nickel, Type K16 for Steel bolt material
P5 = Passivate for CRES bolt material



Examples of part number:

AA60035-4-610QX = a NASM6 - Ø.3750 steel bolt, 1.078 long, 0.438 grip with drilled threads only, Zinc-Nickel plated.
AA60035-4-6CH10P5 = a NASM6 - Ø.3750 CRES bolt, 1.078 long, 0.438 grip with drilled threads and drilled head, passivated.

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AA60035-4-6C10AP5 = a NASM6 - Ø.3750 CRES bolt, 1.078 long, 0.438 grip with undrilled thread and head, passivated.

TABLE V – Type V (-5) - Bolt, self-locking, steel, 160 KSI Ftu, 250 °F, 12 point, external wrenching cap screws.

A-A-60035 Basic Part Number (Alloy Steel, Zinc Nickel Plated)	Length	Reference Dash Number from NASM21098 (MS21098-)					
		0.2500-28 UNF-3A	0.3125-24 UNF-3A	0.3750-24 UNF-3A	0.4375-20 UNF-3A	0.5000-20 UNF-3A	0.6250-18 UNF-3A
		DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
AA60035-5-*QX	0.375	17	18	---	---	---	---
AA60035-5-*QX	0.437	23	24	25	---	---	---
AA60035-5-*QX	0.500	30	31	32	---	---	---
AA60035-5-*QX	0.625	37	38	39	---	---	---
AA60035-5-*QX	0.750	44	45	46	---	---	---
AA60035-5-*QX	0.875	50	51	52	---	---	---
AA60035-5-*QX	1.000	56	57	58	59	60	61
AA60035-5-*QX	1.250	64	65	66	67	68	69
AA60035-5-*QX	1.500	72	73	74	75	76	77
AA60035-5-*QX	1.750	79	80	81	82	83	84
AA60035-5-*QX	2.000	85	86	87	88	89	90
AA60035-5-*QX	2.250	91	92	93	---	94	101
AA60035-5-*QX	2.500	---	95	96	---	97	98
AA60035-5-*QX	3.000	---	---	99	---	100	102

* Add dash number to indicate length.

Example of part number:

AA60035-5-58QX = a NASM21098 – bolt, self-locking, 12 point steel, Zinc-Nickel plated, 0.3750-24 UNF-3A thread, 1 inch long.

AA60035-5-100QX = a NASM21098 – bolt, self-locking, 12 point steel, Zinc-Nickel plated, 0.5000-20 UNF-3A thread, 3 inch long.

TABLE VI – Type VI (-6) - Bolt, tension, steel, external wrenching, flanged, 12-Point, 180 KSI Ftu, 450 ºF.

A-A-60035 Part Number (Alloy Steel, Zinc-Nickel Plated)	Reference Dia. Dash Number (NASM21250)
AA60035-6-02*QX	MS21250-02
AA60035-6-03*QX	MS21250-03
AA60035-6-04*QX	MS21250-04
AA60035-6-05*QX	MS21250-05
AA60035-6-06*QX	MS21250-06
AA60035-6-07*QX	MS21250-07
AA60035-6-08*QX	MS21250-08
AA60035-6-09*QX	MS21250-09
AA60035-6-10*QX	MS21250-10
AA60035-6-12*QX	MS21250-12
AA60035-6-14*QX	MS21250-14
AA60035-6-16*QX	MS21250-16
AA60035-6-18*QX	MS21250-18
AA60035-6-20*QX	MS21250-20
AA60035-6-22*QX	MS21250-22
AA60035-6-24*QX	MS21250-24

* Grip length in sixteenths from NASM21250.

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Examples of part number:

AA60035-6-04008QX = a NASM21250 – Bolt undrilled head, 0.2500-28, 0.500 grip length, Zinc-Nickel plated.
AA60035-6-H04008QX = a NASM21250 – Bolt drilled head, 0.2500-28, 0.500 grip length, Zinc-Nickel plated.

TABLE VII – Type VII (-7) - Bolt, square neck, round head (carriage), steel, Zinc-Nickel plated, UNC-2A

A-A-60035 Part Number (Carbon Steel, Zinc-Nickel Plated)	L Length	Reference Dash Number from NASM35751 (MS35751-)						
		.1900-24 UNC-2A	.2500-20 UNC-2A	.3125-18 UNC-2A	.3750-16 UNC-2A	.5000-13 UNC-2A	.6250-11 UNC-2A	.7500-10 UNC-2A
		Dash No.	Dash No.	Dash No.	Dash No.	Dash No.	Dash No.	Dash No.
AA60035-7-*QX	.750	1	15	40	68	---	---	---
AA60035-7-*QX	.875	211	---	---	---	---	---	---
AA60035-7-*QX	1.000	2	16	41	69	123	---	---
AA60035-7-*QX	1.250	3	17	42	70	124	---	---
AA60035-7-*QX	1.500	4	18	43	71	125	152	---
AA60035-7-*QX	1.750	5	19	44	72	126	153	---
AA60035-7-*QX	2.000	6	20	45	73	127	154	181
AA60035-7-*QX	2.250	7	21	46	74	128	---	182
AA60035-7-*QX	2.500	8	22	47	75	129	156	183
AA60035-7-*QX	2.750	9	23	48	76	130	157	---
AA60035-7-*QX	3.000	10	24	49	77	131	158	185
AA60035-7-*QX	3.250	11	25	50	78	132	159	---
AA60035-7-*QX	3.500	12	26	51	79	133	160	187
AA60035-7-*QX	3.750	13	27	52	80	134	---	---
AA60035-7-*QX	4.000	14	28	53	81	135	162	189
AA60035-7-*QX	4.250	---	202	213	205	209	---	---
AA60035-7-*QX	4.500	---	29	54	82	136	163	190
AA60035-7-*QX	4.750	---	203	204	206	210	---	---
AA60035-7-*QX	5.000	---	30	55	83	137	164	191
AA60035-7-*QX	5.500	---	31	56	84	138	165	192
AA60035-7-*QX	6.000	---	32	57	85	139	166	193
AA60035-7-*QX	6.500	---	---	---	86	140	167	194
AA60035-7-*QX	7.000	---	33	58	87	141	168	195
AA60035-7-*QX	7.500	---	34	59	88	142	169	196
AA60035-7-*QX	8.000	---	35	60	89	143	170	197
AA60035-7-*QX	8.500	---	36	61	90	144	---	198
AA60035-7-*QX	9.000	---	212	63	91	145	172	199
AA60035-7-*QX	9.500	---	---	64	92	146	---	---
AA60035-7-*QX	10.000	---	---	65	93	147	174	---
AA60035-7-*QX	11.000	---	---	---	94	148	175	201
AA60035-7-*QX	11.500	---	---	214	---	---	---	---
AA60035-7-*QX	12.000	---	---	215	95	149	176	---
AA60035-7-*QX	13.000	---	---	216	217	---	---	---
AA60035-7-*QX	14.000	---	---	---	218	---	---	---

* Add dash number to indicate length from NASM35751.

Examples of part number:

AA60035-7-1QX = a NASM35751 –Bolt, square neck, round head (carriage), steel, 0.1900-24 UNC-2A thread, 0.75 length, Zinc-Nickel plated.
AA60035-7-81QX = a NASM35751 –Bolt, square neck, round head (carriage), steel, 0.3750-16 UNC-2A thread, 4.000 length, Zinc-Nickel plated.
AA60035-7-198QX = a NASM35751 –Bolt, square neck, round head (carriage), steel, 0.7500-10 UNC-2A thread, 8.500 length, Zinc-Nickel plated.

7.2 Source of government documents.

7.2.1 FAR: Copies of the Federal Acquisition Regulations are available online from https://www.acquisition.gov/browse/index/far.

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7.2.2 Executive order (EO). Copies of the EO documents are available online from https://www.epa.gov/greeningepa/executive-order-13843.

7.2.3 Military specifications: Copies of military specifications are available online from https://quicksearch.dla.mil.

7.2.4 AIA documents: Copies of Aerospace Industry Association are available online from https://www.aia-aerospace.org/standards/.

7.2.5 SAE documents. Copies of Society of Automotive Engineers documents are available online from https://www.sae.org.

7.3 Ordering data. The contract or order should specify the following:

a. Title, number and date of this CID.
b. Federal stock number, (if available).
c. Material (steel alloy, steel, or aluminum).
d. Part number (PIN).
e. Quantity.

7.4 Key words:

a. Steel
b. Plating
c. Finish
d. Passivate
e. 12 point

MILITARY INTERESTS:

Custodian:
Army – AT
Navy – AS
Air Force – 71
DLA – CC

Review activities:
Army – AR, AV, CR, MI
Navy – AS, CG, MC, SH, YD
Air Force – 11, 70
DLA – GS, GS4, GS7

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
Army – AT

(Project 5306-2022-021)

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